Cost Estimation Research Report

Vepari AI / AI for Micro Enterprise Growth

Prepared from Claude research outputs from the past week

Project	Vepari AI / AI for Micro Enterprise Growth
Prepared for	Startup TN Hackathon
Prepared from	Research from the internet
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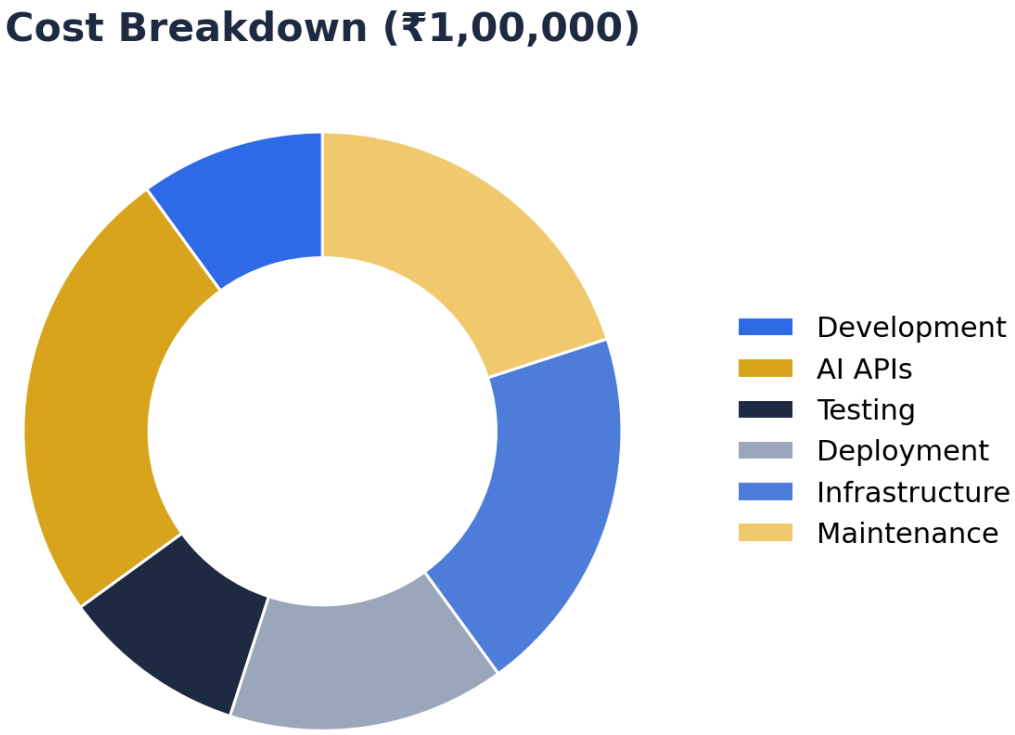
Detailed word / PDF / drive / QR support is provided in the appendices.

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1. Cost Model Summary

The corrected cost model was intentionally reduced from the earlier ₹4.5 lakh-style estimate to a leaner MVP estimate of ₹1,00,000. That reflects the fact that the team is building on existing AI services, not training a foundation model from scratch.



Cost breakdown used in the ₹1,00,000 model

Phase	Estimated cost	Why it exists
Development tools & software	₹10,000	Design tools, repositories, build utilities and core setup
AI API integration & usage	₹25,000	Speech, language and recommendation calls
Testing & quality assurance	₹10,000	Functional checks, bug fixing and voice validation
Deployment & hosting	₹15,000	Launch setup, app hosting and production config
Cloud services & infrastructure	₹20,000	Server-side support, sync and live data access
Maintenance & contingency	₹20,000	Updates, support and unexpected costs

The total is presented as a full MVP build estimate, not a month-by-month burn rate.

2. Development Cost

Cost Driver 1: Development cost is driven by product complexity, not by the number of screens

Observation	The app is not a simple UI wrapper. It needs voice input, local storage, AI interpretation and live intelligence layers.
Evidence	Claude repeatedly framed Vepari AI as a business brain. That means the cost comes from the logic and integration layers, not only from the visible interface.
Analysis	The MVP must support Tamil voice, memory, financial analysis, scheme matching and recommendation logic.
Conclusion	The development budget should therefore cover frontend framework work, backend APIs, storage design and AI integration.
Implication for Vepari AI	This is why development is the largest bucket in the final estimate.

Development item	Cost	Why it is needed
UI / UX design	₹15,000	Premium but clean interface using the same visual language as the PPT
Frontend framework build	₹35,000	Mobile app foundation, screens and state handling
Backend and APIs	₹20,000	Service orchestration, auth, data syncing and logic
Local database architecture	₹10,000	Offline-first storage for business memory
AI integration layer	₹20,000	Voice, NLP and recommendation orchestration
Security / privacy setup	₹0–5,000	Basic encryption and safe storage patterns

3. Testing Cost

Cost Driver 2: Testing is necessary because the app depends on voice, AI and local storage together

Observation	Claude treated noisy-environment voice accuracy and workflow correctness as real risks.
Evidence	The app must be tested for speech recognition quality, UI stability, local-data consistency and recommendation usefulness.
Analysis	If testing is weak, the product can fail even when the core code runs.
Conclusion	The testing budget must include functional QA, prompt / response checks and voice-in-context validation.
Implication for Vepari AI	This is not overhead; it protects the credibility of the AI output.

Testing item	Cost	What it covers
Functional testing	₹4,000	Screen navigation, input-output correctness and crash testing
AI behavior testing	₹3,000	Prompt quality, memory recall and recommendation sanity checks
Voice quality testing	₹3,000	Tamil voice input, speech-to-text and response rendering

4. Production and Deployment Cost

Cost Driver 3: Production cost exists because live AI and live data are external services

Observation	The research repeatedly noted that every user interaction can trigger speech-to-text, text generation, and text-to-speech calls, while market and scheme intelligence require live data retrieval.
Evidence	That means the product has recurring external usage costs even after the software is built.
Analysis	Production spend is needed for API usage, live-server configuration, app deployment and launch reliability.
Conclusion	The deployment budget should therefore include both production setup and the first launch window.
Implication for Vepari AI	This is the cost of making the product real, not just making it exist in code.

Production item	Cost	Why important
AI service calls	₹10,000	Core inference and response generation
Voice service calls	₹7,000	Tamil speech-to-text and response synthesis
Live data access	₹3,000	Schemes, market prices and external intelligence
Deployment / release config	₹15,000	Production build, signing and release pipeline
App hosting / server setup	₹10,000	Launch infrastructure and sync endpoints

The production bucket is deliberately visible because it is usually hidden inside startup estimates. In this project, the live-service cost matters.

5. Maintenance Cost

Cost Driver 4: Maintenance is required because the system is personalized and living

Observation	Claude repeatedly described Vepari AI as a personalized business memory system that gets more useful over time.
Evidence	That kind of product needs upkeep: bug fixes, model tuning, content updates, live data checks and small feature maintenance.
Analysis	A zero-maintenance assumption would be unrealistic for an AI system that handles changing business and scheme data.
Conclusion	The budget should include a monthly support allowance and a contingency buffer for updates and corrections.
Implication for Vepari AI	Maintenance keeps the product credible after launch.

Maintenance item	Monthly / buffer cost	Why it exists
Monitoring and fixes	₹6,000	Bug patches, reliability checks and response review
AI prompt / model tuning	₹5,000	Refinement of the Tamil business brain behavior
Cloud / API buffer	₹6,000	Usage spikes, sync events and live data overhead
Contingency reserve	₹3,000	Unexpected support or rebuild work

6. Why the Cost Is Worth It

The cost is justified because Vepari AI is not a normal mobile app. It is an AI-powered business intelligence system that must continuously translate Tamil voice, store memory locally, query live data and produce personalized guidance.

- AI training / prompt tuning is required to understand Tamil voice, business language, credit patterns and profit patterns.
- Speech-to-text and text-to-speech API calls are repeated in a real product, so every user interaction has a live service cost.
- Live market and scheme intelligence depends on external data sources that must be queried and maintained.
- Local storage for business memory reduces cloud dependence but still needs secure implementation and sync design.
- Long-term business memory means the system accumulates structured data over time, which has storage and maintenance implications.

Live example: one user voice note can trigger speech recognition → AI understanding → local memory update → live scheme lookup → smart recommendation → speech output. That is why the budget is not just “app screens”; it is the cost of the intelligence pipeline.

7. References and Source Index

The links below were explicitly used in the Claude research chain and are included as live verification sources for the cost model.

Cost Model Source Register

This source register lists the verified links explicitly used in the Claude research chain. It is organized by category so evaluators can check the evidence trail quickly.

Category	Source	Link / Use
AI Cost	Sarvam AI pricing / product pages	https://www.sarvam.ai/
AI Infra	Sarvam AI text-to-speech	https://www.sarvam.ai/text-to-speech
AI Infra	Sarvam AI speech-to-text	https://www.sarvam.ai/apis/speech-to-text
Language	Bhashini	https://bhashini.gov.in/
Cloud	Typical cloud-hosting references used in research	https://aws.amazon.com/
Messaging	WhatsApp Business Platform	https://www.whatsapp.com/business/
Market	ThePrint smartphone penetration article	https://theprint.in/economy/rural-areas-will-be-growth-driver-of-internet-smartphone-penetration-in-india-iamai/2234947/

CTA: Use the Drive / QR repository to open the complete cost model, assumptions sheet and external service references.